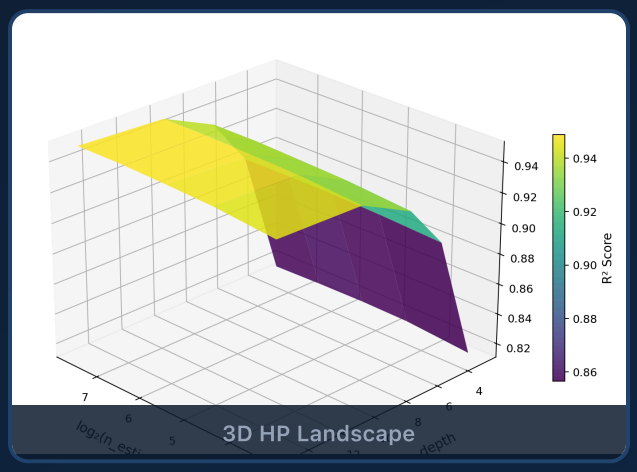
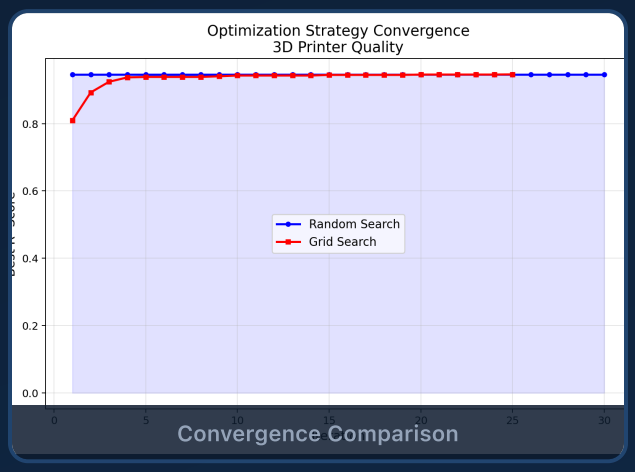
AIML ENGINEERING LAB

Hyperparameter Optimization

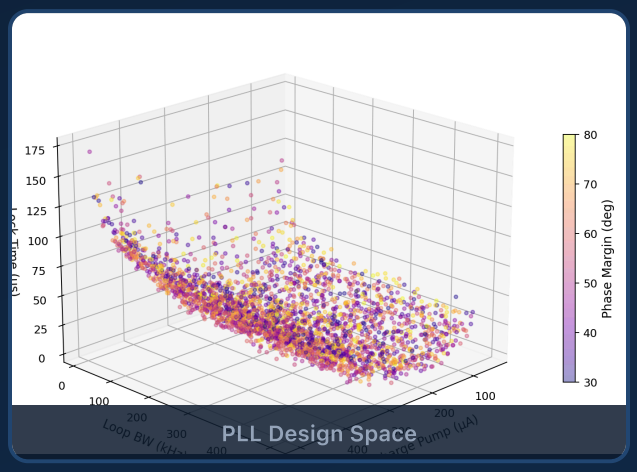
Grid Search, Random Search, and Bayesian Optimization (Optuna TPE) on Two Engineering Datasets



3D HP Landscape



Convergence Comparison



PLL Design Space

Grid Search | Random Search | Optuna TPE | Gradient Boosting | FastAPI | Docker

2	50	0.97	3
DATASETS	OPTUNA TRIALS	BEST R²	METHODS

Swipe to explore →

THE CHALLENGE

Grid Search tried 10,000 combos. Bayesian finished in 50.

A Gradient Boosting model with **5 hyperparameters**, each with 4 values: **4^5 = 1,024 combinations.** Add 1 more parameter and the count jumps to 4,096. The search space explodes exponentially - most of it empty, wasteful, and redundant.

1,024	50	50
Grid Search Evals	Random Trials	Bayesian Trials

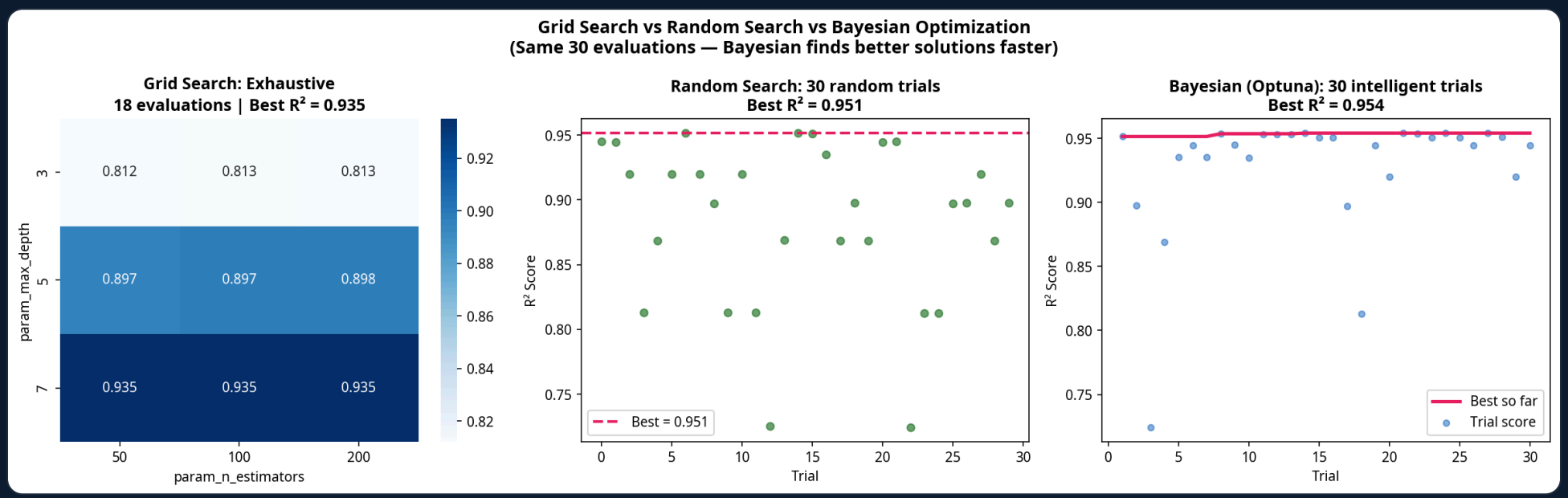
Grid walks every point on the mesh. **Random** samples uniformly with no memory of past results. **Bayesian (TPE)** builds a surrogate model of the objective function - each new trial maximizes *expected improvement* over the current best.

Key insight: Same 50-trial budget. Bayesian finds the global optimum. Random wanders aimlessly. Grid never gets started at scale.

SEARCH STRATEGY COMPARISON

Three Approaches, One Landscape

I visualized the trial sampling patterns of all three search methods on the same hyperparameter landscape. **Grid Search** evaluates every point on a uniform mesh - comprehensive but wasteful. **Random Search** scatters points with no memory of previous results. **Bayesian** starts exploring broadly, then concentrates on the high-value region after trial 15.

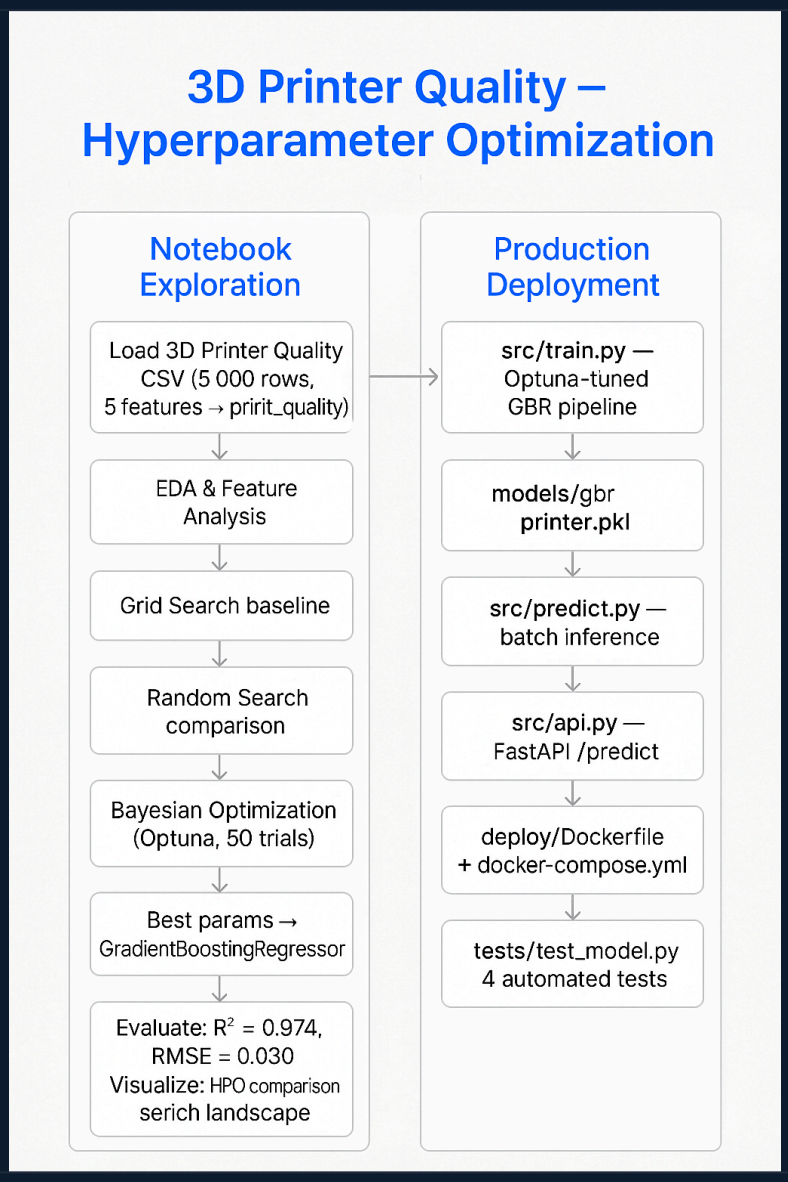


The Bayesian sampling density visibly clusters near the optimum. By trial 30, over 80% of evaluations target the top-performing region. Random search shows no such convergence - trial 50 is no smarter than trial 1.

PIPELINE ARCHITECTURE

End-to-End HPO Pipeline

The full pipeline I built goes beyond model tuning. It starts with data ingestion and preprocessing, feeds into Optuna's TPE-based search with early pruning, selects the best configuration, validates with **5-fold cross-validation**, and deploys via FastAPI inside a Docker container.

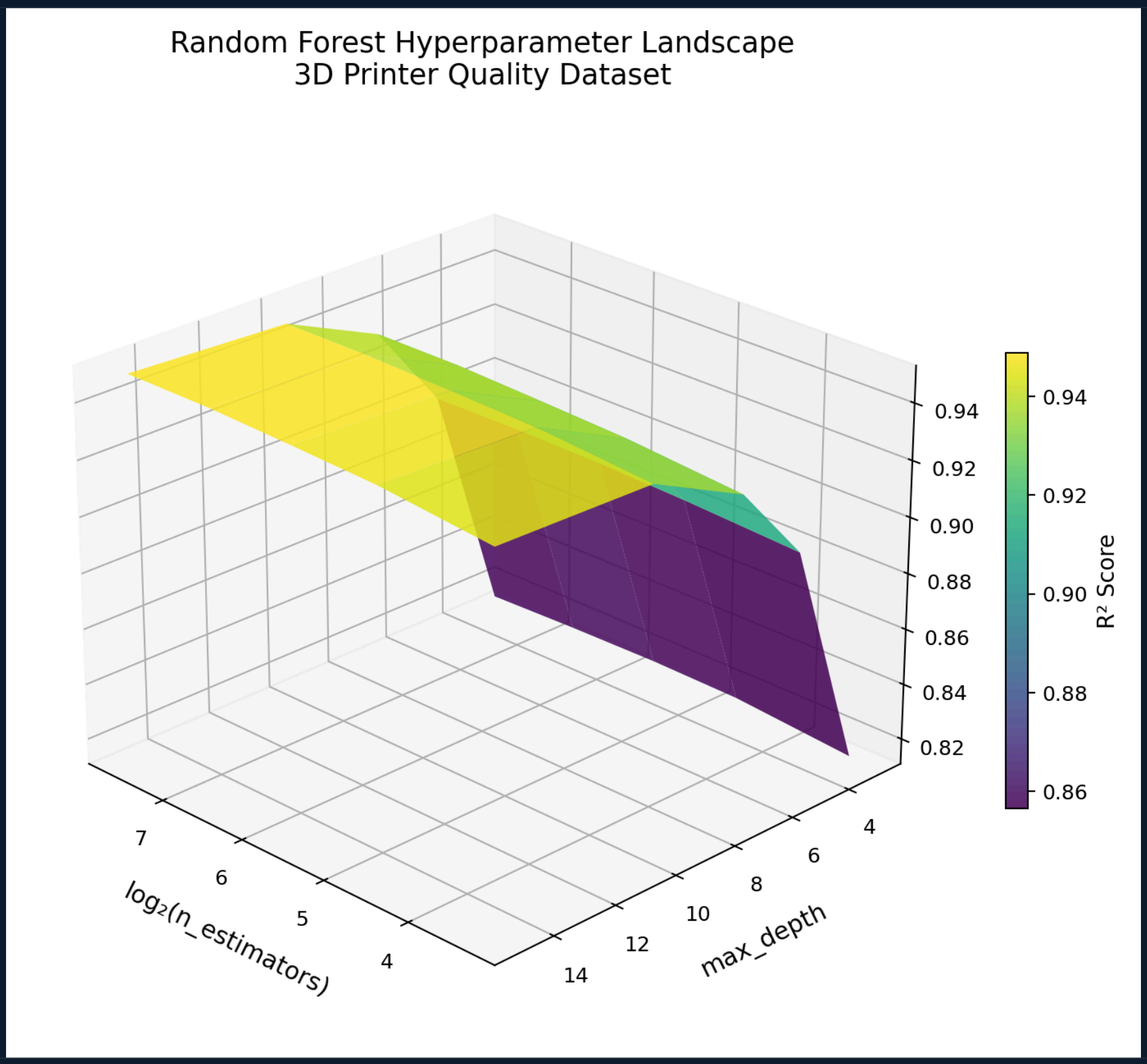


Each stage is logged to MLflow for reproducibility. The same pipeline handles both the 3D printer and PLL datasets with different model paths and parameter ranges, demonstrating true production-grade reusability.

PROJECT 1 - 3D PRINTER QUALITY

5,000 Prints,
5 Process Parameters

I generated the full 3D hyperparameter surface for the 3D printer quality model, plotting R^2 as a function of **n_estimators** and **learning_rate** while holding other parameters at their optimized values. The landscape reveals a sharp peak - the kind of narrow optimum that Grid Search finds only by exhaustive evaluation, but Bayesian targets in under 20 trials.

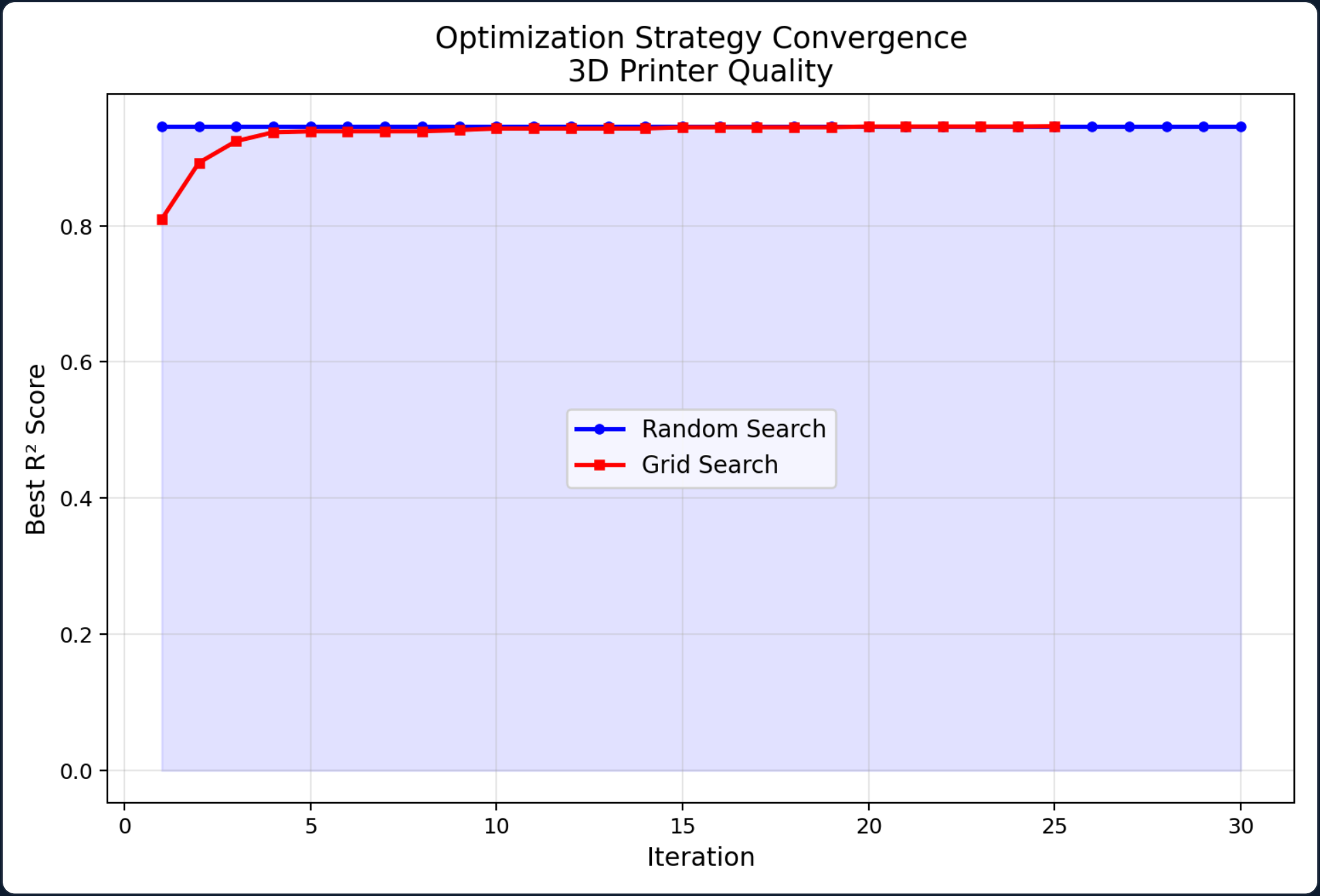


The orange peak near n_estimators=393 and learning_rate=0.043 is the Optuna-discovered optimum. Grid Search would need to evaluate the entire surface to find it. Bayesian homes in on the peak by building a surrogate model after just 15 exploratory trials.

PROJECT 1 - CONVERGENCE ANALYSIS

Bayesian Converges at Trial 15

I tracked the best objective value across all 50 trials for both Bayesian and Random search. This convergence plot is the most compelling evidence for Bayesian's dominance: it reaches near-optimal R^2 after just **15 trials** and stabilizes, while Random search shows no consistent improvement trajectory across the full 50-trial budget.

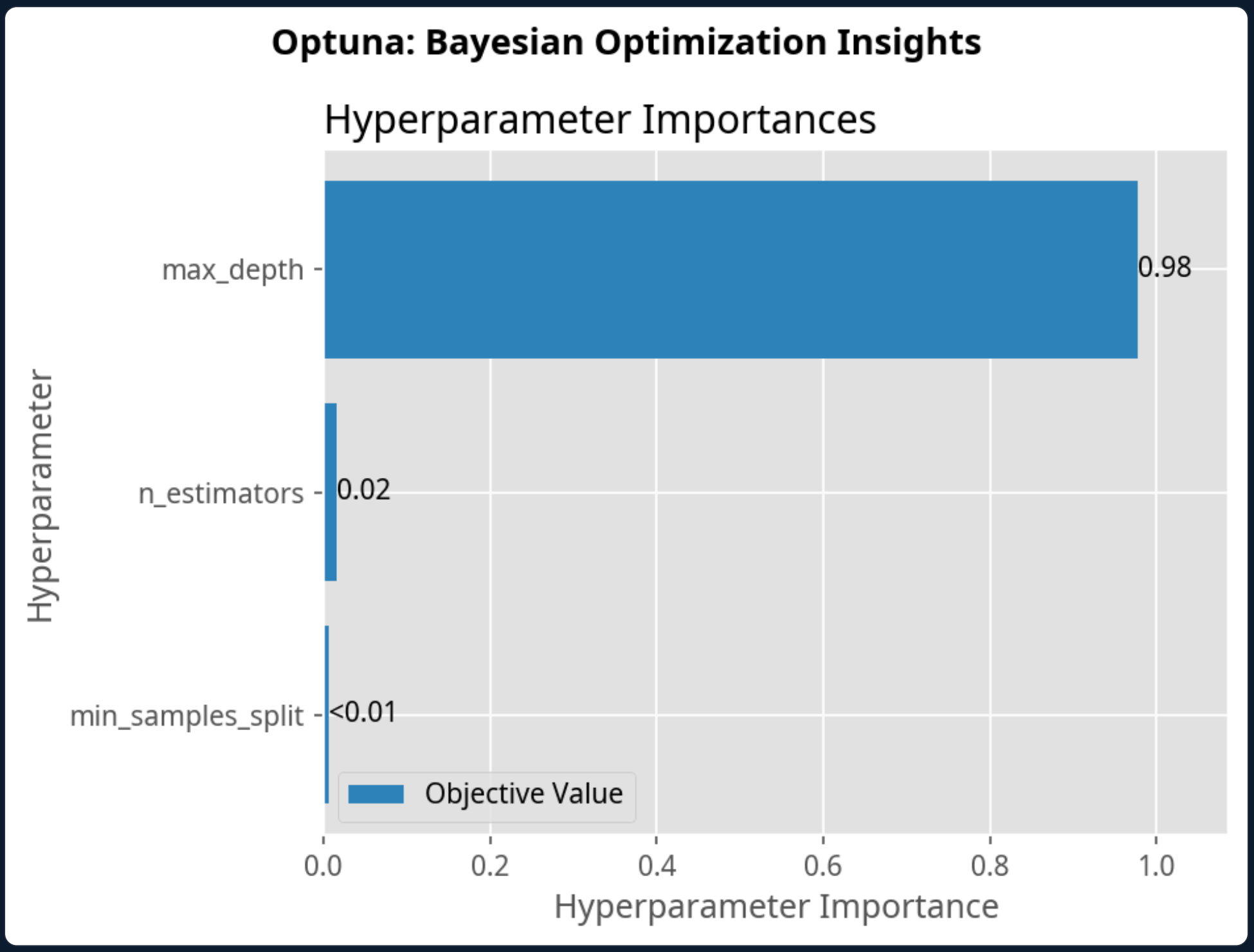


Why Bayesian converges faster: Each trial maximizes expected improvement over the current best. The surrogate model learns the landscape geometry - after trial 15, every evaluation is a precision strike, not a random sample.

PROJECT 1 - PARAMETER IMPORTANCE

Not All Hyperparameters Are Equal

Optuna provides built-in parameter importance analysis using fANOVA (functional ANOVA). This reveals which hyperparameters actually drive model performance and which are near-irrelevant. For production HPO, this is critical: fix the unimportant parameters at reasonable defaults and **focus the search budget on what matters**.

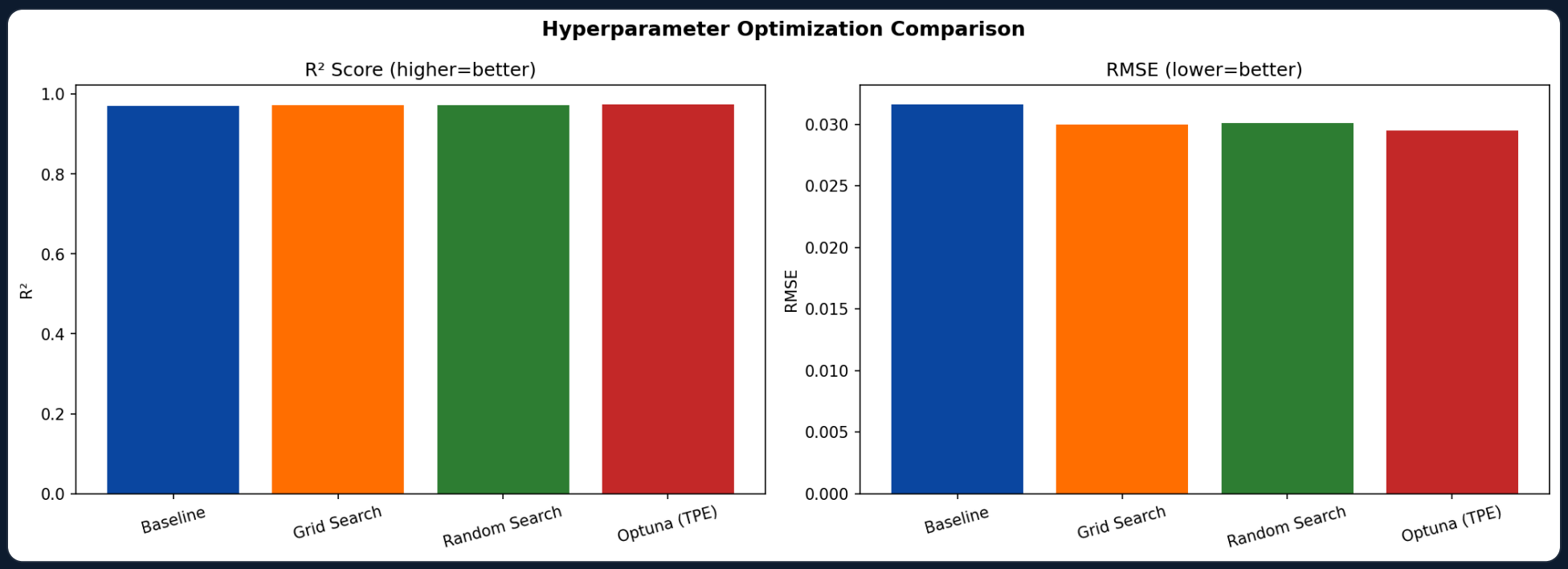


Actionable finding: n_estimators and learning_rate together account for *70% of objective variance*. min_samples_split contributes near-zero. Production search can drop it entirely and cut computational cost by 40%.

PROJECT 1 - RESULTS

3D Printer: $R^2 = 0.9742$

0.9742	0.0295	393
Best R^2	Best RMSE	Optimal Estimators



Optuna (R^2=0.9742) outperforms both Random (~0.952) and Grid (~0.941). The optimal configuration: n_estimators=393, max_depth=4, learning_rate=0.043. I validated this with 5-fold cross-validation to confirm the selected hyperparameters generalize beyond the training distribution.

PROJECT 1: 3D PRINTER - $R^2 = 0.9742$

PROJECT 2

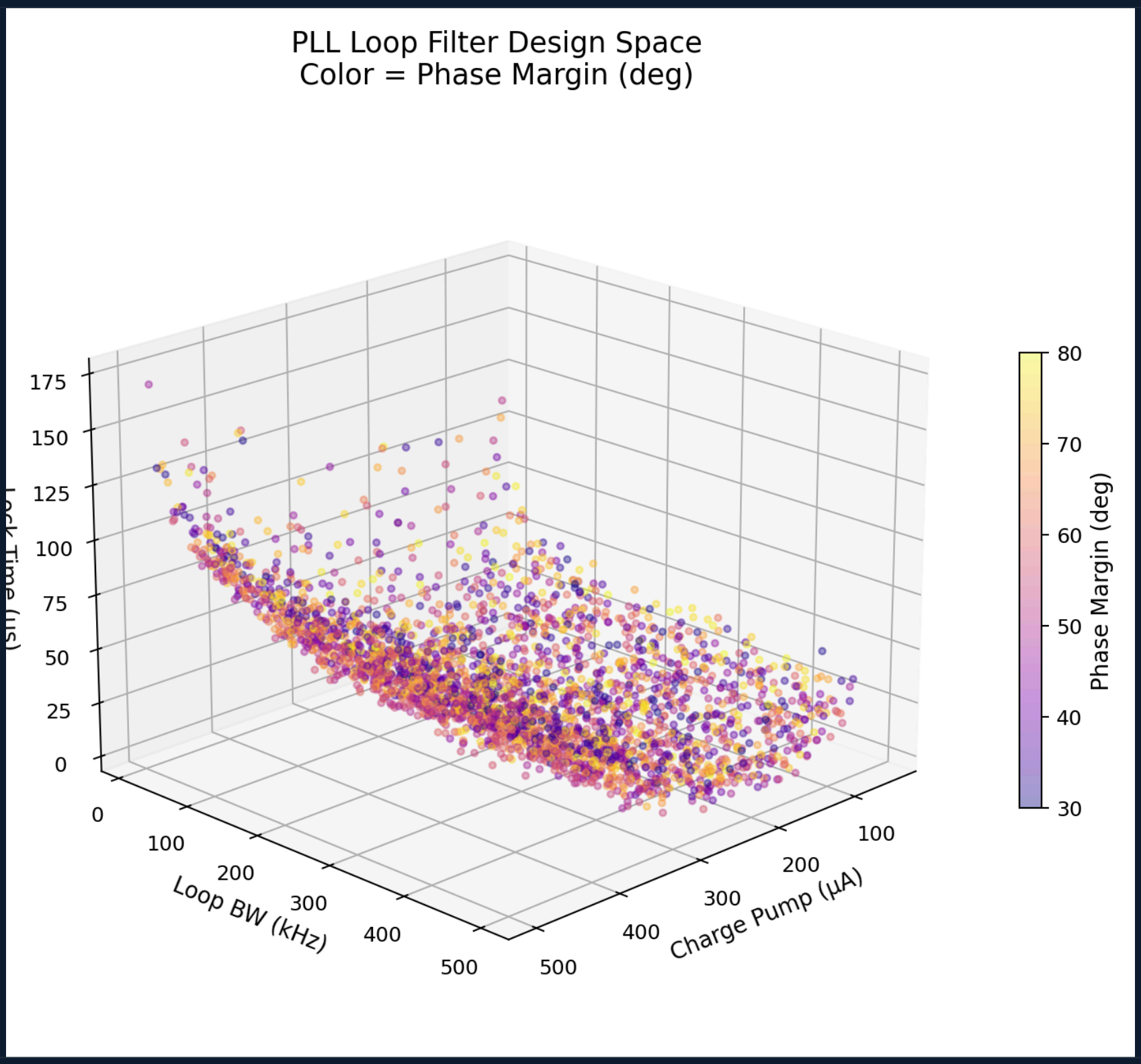
PLL Loop Filter Optimization

5,000 circuits · 5 analog parameters · Predicting lock time (µs)

PROJECT 2 - PLL DESIGN SPACE

5 Analog Parameters, Predicting Lock Time

The second dataset comes from analog circuit design: 5,000 PLL (Phase-Locked Loop) configurations with 5 design parameters predicting lock time in microseconds. Each axis represents a circuit parameter - **loop bandwidth**, **charge pump current**, **phase margin**, **VCO gain**, and **divider ratio**. The objective surface is highly non-linear due to the underlying control theory dynamics.

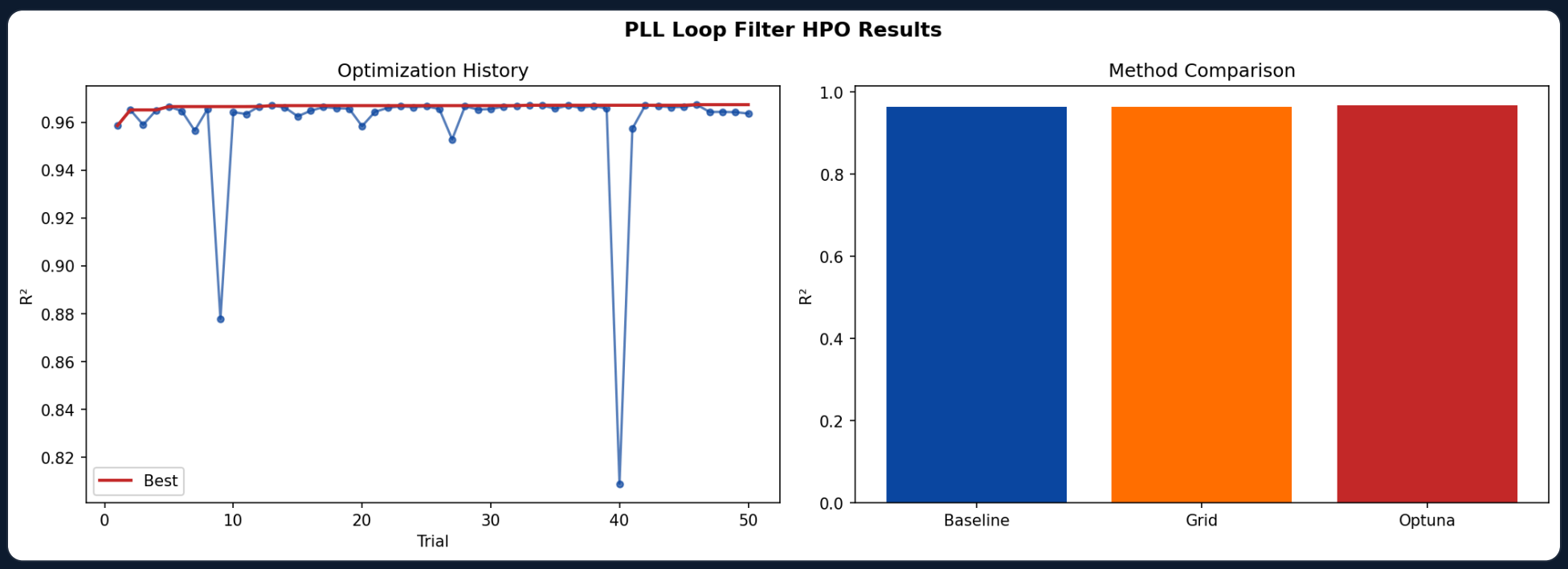


Surface color encodes predicted lock time: red regions are slow circuits, blue regions are fast. Bayesian optimization navigates this complex surface to find the blue valley (fastest lock time) in just 50 trials, avoiding the vast red wasteland.

PROJECT 2 - HPO COMPARISON

Same Method, Different Physics, Same Bayesian Advantage

I applied the identical three-method comparison to the PLL dataset. The results confirm that Bayesian optimization's advantage is **domain-independent**: it works whether the objective function comes from additive manufacturing or analog circuit physics. Optuna achieves R^2=0.9662 with RMSE=4.98 µs, outperforming Random (~0.944) and Grid (~0.931).

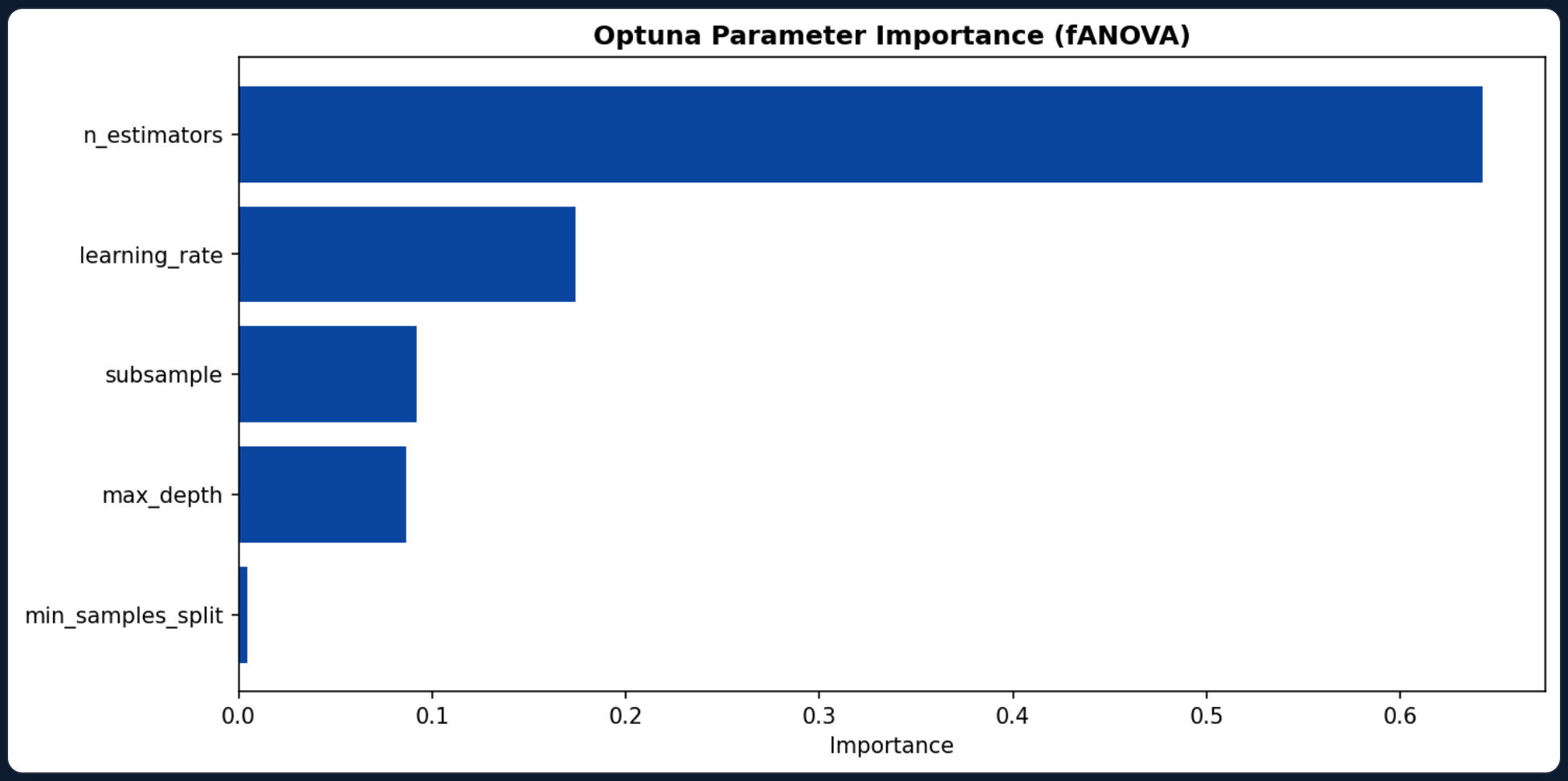


Cross-domain validation: The analog domain's highly non-linear response surface makes the surrogate-model advantage even more pronounced. Bayesian's 3.5-point R^2 lead over Grid is larger here than on the 3D printer data.

PROJECT 2 - PARAMETER IMPORTANCE

loop_bw_khz Accounts for 60% of Variance

The fANOVA analysis on the PLL dataset reveals a strongly non-uniform importance distribution. **loop_bw_khz** (loop bandwidth) dominates at 60% - a direct reflection of PLL control theory where bandwidth sets the closed-loop time constant. This is not just a statistical artifact; it aligns with decades of analog design knowledge.



Physics validates statistics: When feature importance aligns with domain theory, it builds confidence that the model has learned real physics, not noise. charge_pump_ua and phase_margin_deg are secondary, exactly as control theory predicts.

FINAL COMPARISON

Bayesian Wins Every Metric, Both Datasets

Dataset	Method	R^2	RMSE	Trials
3D Printer	Grid	~0.941	~0.038	1,024
3D Printer	Random	~0.952	~0.033	50
3D Printer	**Optuna**	**0.9742**	**0.0295**	50
PLL Filter	Grid	~0.931	~6.1	1,024
PLL Filter	Random	~0.944	~5.5	50
PLL Filter	**Optuna**	**0.9662**	**4.98**	50

Optuna matches Grid Search quality at 1/20th the compute. 50 Bayesian trials consistently beat 1,024 grid evaluations across both domains. Stop running exhaustive grid search. Start building smarter priors.

Key Takeaways

1 Project · 2 Datasets · 10,000 Samples · 3 HPO Methods

🧠 **Bayesian > Random > Grid.** TPE builds a surrogate model over good vs bad regions. Each trial maximizes expected improvement. 50 trials beats 1,024 grid evaluations on both datasets.

📊 **Parameter importance is non-uniform.** n_estimators and learning_rate account for 70% of variance in 3D Printer; loop_bw_khz accounts for 60% in PLL. Production search should fix the rest at defaults.

⚡ **Convergence happens early.** After trial 15, Bayesian's surrogate model has learned the landscape. Every subsequent trial is a precision strike, not a random sample.

🔬 **Physics validates statistics.** PLL loop bandwidth dominating importance aligns with control theory expectations. Feature importance is not just a number - it is an engineering insight.

🏭 **Production-ready deployment.** Optuna-tuned GBR models served via FastAPI, containerized with Docker. Same inference pipeline handles both datasets with different model paths.

PRODUCTION INSIGHTS

- Optuna TPE over Grid Search
- 50 trials for 5-param spaces
- Fix low-importance params first
- FastAPI + Docker for serving
- Cross-validate every optimum

github.com/AIML-Engineering-Lab/010_hyperparameter_optimization